Table 1: Controller performance data under free vibration conditions. The mean maximum amplitude is expressed in metres. The TD3 and GRUA-TD3 rows each summarise three independent training runs, every run contributing the checkpoint with the highest mean reward over a 20-episode window of its training curve; each checkpoint is evaluated over ten random realisations of the loop delay and of the sampling jitter that are shared with the corresponding no-control response, so that every value is the mean ± standard deviation over thirty episodes.

Control Strategy	Ideal Environment		Non-Ideal Environment Delay-(3,1)	
	Mean Max. Amplitude (m)	Attenuation Rate	Mean Max. Amplitude (m)	Attenuation Rate
No Control	4.60×10^{-3}	0.00%	4.60×10^{-3}	0.00%
PID	7.99×10^{-4}	82.64%	2.44×10^{-3}	46.91%
TD3	$(1.66 \pm 0.47) \times 10^{-3}$	$63.90 \pm 10.19\%$	$(1.96 \pm 1.31) \times 10^{-3}$	$57.41 \pm 28.42\%$
GRUA-TD3	$(7.80 \pm 2.92) \times 10^{-4}$	$83.05 \pm 6.34\%$	$(8.80 \pm 4.92) \times 10^{-4}$	$80.88 \pm 10.70\%$

Table 2: Controller performance data under fixed-frequency excitation conditions. The fixed-frequency excitation is a 30 Hz sinusoid of amplitude $F_0 = 150\,\mathrm{N}$, and the mean maximum amplitude is expressed in metres. The TD3 and GRUA-TD3 rows each summarise three independent training runs, every run contributing the checkpoint with the highest mean reward over a 20-episode window of its training curve; each checkpoint is evaluated over ten random realisations of the loop delay and of the sampling jitter that are shared with the corresponding no-control response, so that every value is the mean ± standard deviation over thirty episodes.

Control Strategy	Ideal Environment		Non-Ideal Environment Delay-(3,1)	
	Mean Max. Amplitude (m)	Attenuation Rate	Mean Max. Amplitude (m)	Attenuation Rate
No Control	6.50×10^{-3}	0.00%	6.12×10^{-3}	0.00%
PID	1.18×10^{-3}	81.85%	2.32×10^{-3}	62.02%
TD3	$(1.61 \pm 0.25) \times 10^{-3}$	$72.28 \pm 4.83\%$	$(2.18 \pm 0.22) \times 10^{-3}$	$62.39 \pm 5.09\%$
GRUA-TD3	$(1.03 \pm 0.18) \times 10^{-3}$	$82.20 \pm 4.12\%$	$(8.06 \pm 1.19) \times 10^{-4}$	$86.08 \pm 2.48\%$

Table 3: Controller performance data under random excitation conditions. The excitation is an MLS random signal approximated by a periodic pseudo-random binary sequence, and the mean maximum amplitude is expressed in metres. The TD3 and GRUA-TD3 rows each summarise three independent training runs, every run contributing the checkpoint with the highest mean reward over a 20-episode window of its training curve; each checkpoint is evaluated over ten random realisations of the loop delay and of the sampling jitter that are shared with the corresponding no-control response, so that every value is the mean ± standard deviation over thirty episodes.

Control Strategy	Ideal Environment		Non-Ideal Environment Delay-(3,1)	
	Mean Max. Amplitude (m)	Attenuation Rate	Mean Max. Amplitude (m)	Attenuation Rate
No Control	9.50×10^{-3}	0.00%	9.21×10^{-3}	0.00%
PID	5.40×10^{-3}	43.14%	2.60×10^{-3}	71.72%
TD3	$(1.90 \pm 0.32) \times 10^{-3}$	$79.99 \pm 3.41\%$	$(5.04 \pm 2.94) \times 10^{-3}$	$46.93 \pm 31.01\%$
GRUA-TD3	$(1.75 \pm 0.69) \times 10^{-3}$	$81.59 \pm 7.05\%$	$(2.11 \pm 0.60) \times 10^{-3}$	$77.77 \pm 6.27\%$